

GOD'S EYE

Integrated Teacher vs Student Baseline Evaluation

Research stage: Pre-distillation baseline

This report documents the corrected same-frame evaluation of the two baseline systems: **Teacher = YOLO11-L + DPT-Large** and **Student = YOLO11-S + DPT-Small**. The Distilled Student has not yet been trained or evaluated.

The objective is to establish a clean baseline for the later knowledge-distillation experiment: measure how much task performance is lost by moving to the compact Student, and how much of that gap can subsequently be recovered through distillation.

Evaluation datasets: KITTI and NYU Depth V2

Evaluation design: fixed same-frame manifests, shared thresholds and calibration protocol, batch size 1

Evaluation status: Teacher and Student complete; Distilled Student pending

1. Executive Summary

The baseline results show a clear accuracy-efficiency trade-off. The Student is substantially cheaper and faster than the Teacher on both datasets, while its task-performance loss is dataset dependent. On KITTI, the Student loses detection mAP, depth quality, and object-distance accuracy. On NYU, the Student's depth quality is nearly unchanged and its object-distance error is actually lower than the Teacher's, although detection metrics remain lower.

Main finding: the Student is not simply a uniformly weaker model. It is a highly efficient alternative with different strengths and weaknesses. This makes the later distillation experiment meaningful: the goal is not merely to make the Student faster, but to determine whether Teacher knowledge can recover the Student's lost KITTI and other task-specific performance without giving up most of its efficiency advantage.

Key metric	KITTI Teacher	KITTI Student	NYU Teacher	NYU Student
mAP@50	0.336	0.290	0.0237	0.0195
Depth RMSE	8.49 m	10.71 m	0.872 m	0.873 m
Object distance MAE	6.25 m	8.08 m	0.663 m	0.516 m
FPS	2.20	17.87	6.24	14.48
GPU memory	2893 MB	204 MB	2524 MB	220 MB

Interpretation

On KITTI, Student FPS rises from 2.20 to 17.87 frames/s, while GPU memory falls from 2893 MB to 204 MB. On NYU, FPS rises from 6.24 to 14.48. These efficiency gains are the principal benefit of the compact architecture.

2. Evaluation Methodology

Both systems were evaluated as complete GOD'S EYE pairs rather than as independent detector and depth benchmarks. For each RGB frame, the detector and monocular depth model operate on the same image. Detections are matched to ground-truth objects using the shared class-aware IoU protocol, and the predicted depth within the detected object region is converted into an object-distance estimate.

Teacher: YOLO11-L + DPT-Large

Student: YOLO11-S + DPT-Small

Batch size: 1

Evaluation: 500 fixed frames per dataset

Calibration: 100 disjoint calibration frames per dataset

Distilled Student: not yet available

Metrics

Detection: precision, recall, F1, mean IoU, mAP@50, mAP@50-95, matched objects, missed objects, false detections.

Depth: RMSE, AbsRel, SqRel, δ_1 , δ_2 , δ_3 , plus KITTI inverse-depth metrics where applicable.

Object distance: MAE, RMSE, median absolute error, relative error, tolerance accuracy, and end-to-end success rates.

Runtime: FPS, mean/median/P95 latency, latency standard deviation, and stage-level timing.

Resources: CPU, RAM, GPU utilization, GPU memory, parameter count, and model size.

3. KITTI Results

Detection metric	Teacher	Student	Student – Teacher
Precision	0.349	0.377	+0.029
Recall	0.837	0.819	-0.017
F1	0.492	0.517	+0.025
Mean IoU	0.818	0.808	-0.010
mAP@50	0.336	0.290	-0.046
mAP@50-95	0.197	0.167	-0.030

Depth metric	Teacher	Student	Student – Teacher
RMSE (m)	8.49	10.71	+2.23
AbsRel	0.449	0.611	+0.162
SqRel	3.773	6.031	+2.257
δ_1	0.334	0.235	-0.098
δ_2	0.638	0.483	-0.154
δ_3	0.818	0.704	-0.115

KITTI interpretation

The Student shows a clear efficiency advantage but a measurable performance loss on KITTI. Its mAP@50 falls from 0.336 to 0.290, while depth RMSE increases from 8.49 m to 10.71 m. The object-distance MAE also increases from 6.25 m to 8.08 m. This is the clearest evidence in the baseline that there is useful Teacher capability for distillation to recover.

Object-distance metric	Teacher	Student
MAE	6.25 m	8.08 m
RMSE	7.89 m	10.63 m
Median AE	5.49 m	6.18 m
Mean relative error	33.4%	38.2%
Within ± 0.5 m	4.7%	3.8%
Within ± 1.0 m	9.2%	7.4%
End-to-end success ≤ 0.5 m	3.9%	3.1%
End-to-end success ≤ 1.0 m	7.7%	6.1%

4. NYU Depth V2 Results

Detection metric	Teacher	Student	Student – Teacher
Precision	0.057	0.055	-0.002
Recall	0.461	0.394	-0.067
F1	0.101	0.097	-0.004
Mean IoU	0.820	0.809	-0.011
mAP@50	0.0237	0.0195	-0.0042
mAP@50-95	0.0147	0.0108	-0.0038

Depth metric	Teacher	Student	Student – Teacher
RMSE (m)	0.872	0.873	+0.000
AbsRel	0.304	0.302	-0.002
SqRel	0.338	0.340	+0.002
δ_1	0.505	0.524	+0.019
δ_2	0.802	0.800	-0.002
δ_3	0.922	0.920	-0.002

NYU interpretation

The NYU results are substantially less negative for the Student than KITTI. Depth RMSE is effectively unchanged (0.872 m vs 0.873 m), while AbsRel is slightly lower for the Student (0.302 vs 0.304). Object-distance MAE improves from 0.663 m to 0.516 m. Detection mAP remains lower for the Student, so the compact model is not uniformly better, but it demonstrates that substantial compression is possible without a corresponding depth-quality collapse on this dataset.

Object-distance metric	Teacher	Student
MAE	0.663 m	0.516 m
RMSE	0.894 m	0.715 m
Median AE	0.507 m	0.385 m
Mean relative error	22.0%	19.8%
Within ± 0.5 m	49.7%	62.1%
Within ± 1.0 m	80.0%	87.9%
End-to-end success ≤ 0.5 m	21.8%	23.3%
End-to-end success ≤ 1.0 m	35.2%	33.0%

5. Efficiency and Deployment Analysis

Metric	KITTI Teacher	KITTI Student	NYU Teacher	NYU Student
FPS	2.20	17.87	6.24	14.48
Mean latency	454.4 ms	56.0 ms	160.3 ms	69.1 ms
P95 latency	465.0 ms	72.5 ms	166.1 ms	77.8 ms
GPU memory	2893 MB	204 MB	2524 MB	220 MB
Parameters	369.4M	30.8M	369.4M	30.8M
Model size	1361.6 MB	100.2 MB	1361.6 MB	100.2 MB

Key efficiency ratios

On KITTI, the Student is approximately **8.1× faster**, with about **93.0% less peak GPU memory** and **91.7% fewer parameters**. On NYU, it is approximately **2.3× faster** and uses about **91.3% less peak GPU memory**.

The runtime data also shows that DPT-Large dominates Teacher latency, particularly on KITTI. Teacher DPT inference is 406.7 ms/frame on KITTI versus 20.9 ms for DPT-Small. This is the primary computational reason the compact depth model provides a large end-to-end speedup.

6. What the Baseline Means for Knowledge Distillation

The baseline establishes a non-trivial compression problem. The Student already delivers a major computational advantage, so the purpose of distillation is not simply to make inference faster. The Student is already the fast model. The research question is whether Teacher knowledge can improve the Student's task performance without substantially sacrificing that efficiency.

KITTI provides the strongest motivation for distillation. The Student loses mAP, depth quality, and object-distance accuracy relative to the Teacher. The largest gaps are in depth and long-range object-distance performance.

NYU provides a useful counterpoint. The Student's depth metrics are very close to the Teacher's and its object-distance MAE is lower. Therefore, distillation should be evaluated as a hypothesis rather than assumed to improve every metric. A useful result could be selective recovery of the KITTI performance gap while retaining the Student's computational profile.

Planned three-system experiment

System	Detection	Depth	Purpose
Teacher	YOLO11-L	DPT-Large	High-capacity reference
Student	YOLO11-S	DPT-Small	Compact baseline
Distilled Student	Distilled YOLO11-S	Distilled DPT-Small	Test recovery of Teacher knowledge

The final comparison should focus on three quantities: (1) accuracy retained by the Student, (2) accuracy recovered after distillation, and (3) computational efficiency retained after distillation.

7. Limitations and Next Steps

- 1. The Distilled Student has not yet been evaluated.** The current report is therefore a baseline report, not the final knowledge-distillation result.
- 2. Dataset differences matter.** KITTI is an outdoor driving dataset with wide images and sparse LiDAR depth, while NYU is an indoor RGB-D dataset with dense metric depth. Performance should therefore be interpreted primarily within each dataset rather than as a direct cross-dataset ranking.
- 3. Object-distance metrics depend on detection matching.** A distance estimate is only evaluated for matched objects under the defined matching protocol. This makes the end-to-end success metrics particularly important.
- 4. The current results establish baseline behavior, not causal explanations.** The observed KITTI gap may arise from model capacity, dataset characteristics, preprocessing, depth-ground-truth sparsity, or their interaction. The distillation experiment should therefore be accompanied by the same fixed evaluation protocol.

Immediate next steps

1. Lock the Teacher and Student baseline results.
2. Verify and record the exact preprocessing and model configurations used for reproducibility.
3. Train YOLO11-S using the selected distillation objective from YOLO11-L.
4. Train DPT-Small using the selected distillation objective from DPT-Large.
5. Evaluate the Distilled Student on the exact same 500 KITTI and 500 NYU frames.
6. Compare Teacher → Student → Distilled Student for detection, depth, object distance, runtime and resource usage.

Conclusion

The baseline demonstrates that the compact Student architecture offers a very large efficiency advantage, but the trade-off is dataset and task dependent. KITTI exposes clear performance losses in detection, depth and object distance, while NYU shows that depth and object-distance performance can remain competitive despite substantial compression. This establishes a strong and defensible motivation for the next stage: testing whether knowledge distillation can recover the Teacher's more demanding capabilities while preserving the Student's speed and memory advantages.

Appendix A. Baseline Comparison Matrix

Metric	KITTI T	KITTI S	NYU T	NYU S
Precision	34.9%	37.7%	5.7%	5.5%
Recall	83.7%	81.9%	46.1%	39.4%
F1	49.2%	51.7%	10.1%	9.7%
mAP@50	33.6%	29.0%	2.4%	1.9%
mAP@50-95	19.7%	16.7%	1.5%	1.1%
Depth RMSE	8.485 m	10.712 m	0.872 m	0.873 m
Depth AbsRel	44.9%	61.1%	30.4%	30.2%
Depth $\delta 1$	33.4%	23.5%	50.5%	52.4%
Distance MAE	6.250 m	8.080 m	0.663 m	0.516 m
Distance RMSE	7.889 m	10.628 m	0.894 m	0.715 m
Distance rel. error	33.4%	38.2%	22.0%	19.8%
Distance $\leq 0.5\text{m}$	4.7%	3.8%	49.7%	62.1%
Distance $\leq 1.0\text{m}$	9.2%	7.4%	80.0%	87.9%
End-to-end success $\leq 0.5\text{m}$	3.9%	3.1%	21.8%	23.3%
End-to-end success $\leq 1.0\text{m}$	7.7%	6.1%	35.2%	33.0%
FPS	2.20	17.87	6.24	14.48
Mean latency (ms)	454.4	56.0	160.3	69.1
P95 latency (ms)	465.0	72.5	166.1	77.8
GPU memory (MB)	2893.2	203.5	2524.2	220.3
CPU (%)	99.1%	94.8%	99.6%	98.3%
RAM (MB)	3387.6	1463.2	1457.5	1477.7
Parameters	369.4M	30.8M	369.4M	30.8M
Model size (MB)	1361.6	100.2	1361.6	100.2